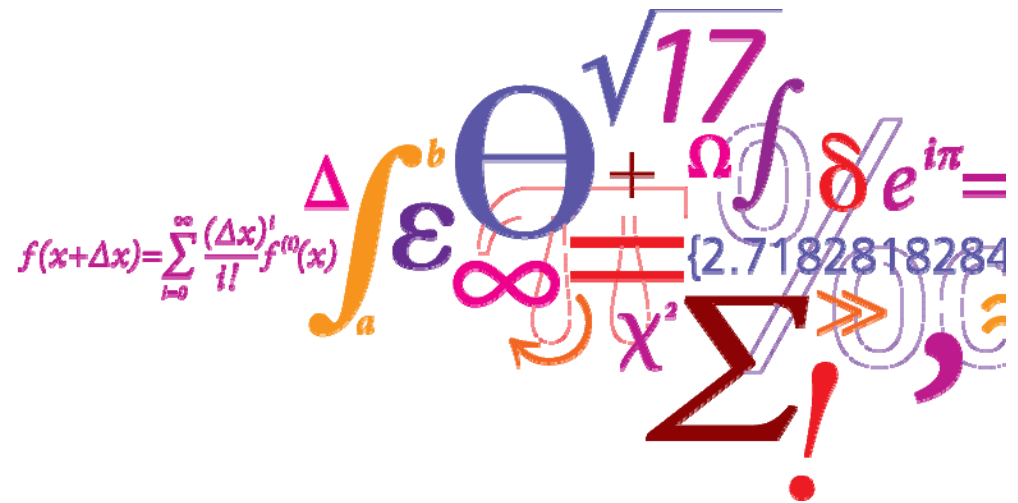


31792 -- Advanced Optimization and Game Theory for Energy Systems

Lecture 3: Desirable properties of market-clearing mechanisms

Jalal Kazempour

January 6, 2021



Recap

- Optimization form of the market-clearing problem maximizes market's social welfare.

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 - Optimization and equilibrium forms of the market-clearing problem are “equivalent”, since their corresponding KKT conditions are identical.
 - Therefore, the market's social welfare is maximized, while each market player obtains its maximum objective.
- This means that the market-clearing mechanism is **efficient**.
- A market fails (market failure) if for any reason, social welfare is not maximized, or there is any player who desires to unilaterally deviate from market-clearing outcomes.

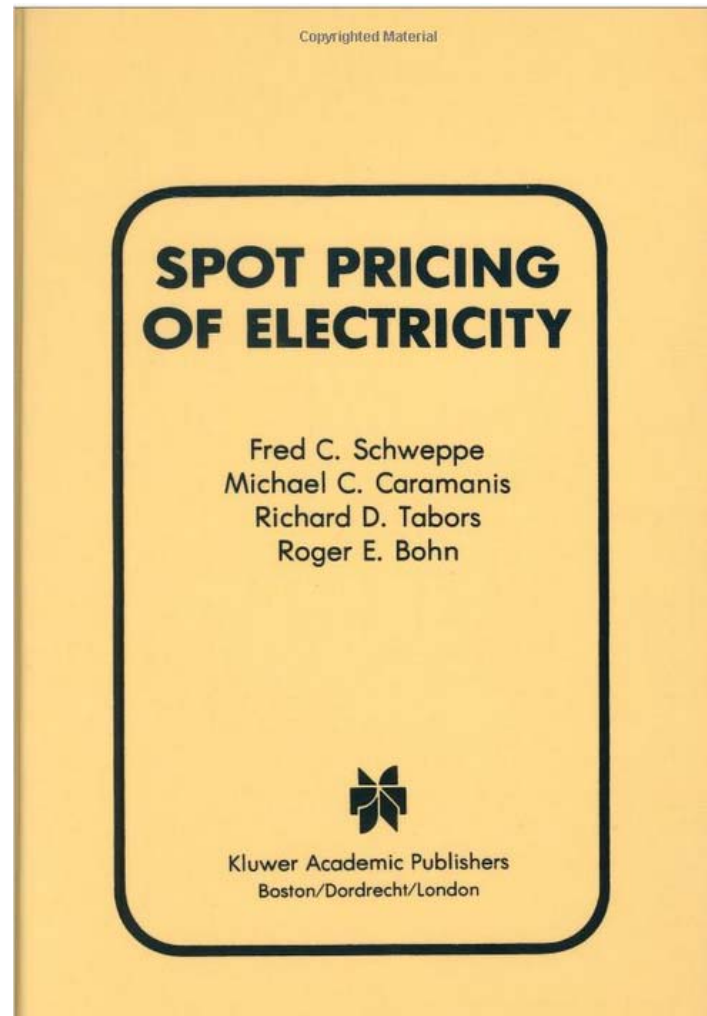
Discussion



Question:

In addition to achieving **market efficiency**, is there any other desirable economic property for a market-clearing mechanism?

A relevant and very nice book!



Desirable economic properties

Four desirable properties of market-clearing mechanisms are:

Market efficiency

Incentive compatibility

Cost recovery

Revenue adequacy

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In an **efficient** market, the social welfare is maximized, and no one desires to unilaterally deviate from the market outcomes.

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Four desirable properties of market-clearing mechanisms are:

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In an **incentive-compatible** market:

- Every market player can maximize its objective just by acting according to her “true” preferences.

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In an **incentive-compatible** market:

- Every market player can maximize its objective just by acting according to her “true” preferences.
- In an incentive-compatible market, if the production cost of a generator is \$12/MWh, the **dominant** (most profitable) strategy for that generator is to offer “trustfully” at \$12/MWh, not at any different price!

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- In other words, no market player desires to exercise “**market power**” by behaving “**strategically**”, i.e., by submitting “strategic” offers.

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- **Cost recovery** refers to a condition under which every market player is able to recover her operational (but not necessarily capital) cost. In other words, her operational profit is always non-negative.

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- **Cost recovery** refers to a condition under which every market player is able to recover her operational (but not necessarily capital) cost. In other words, her operational profit is always non-negative.
- This property is also known as “**individual rationality**” (although based on some definition in the literature there might be slight differences).

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Four desirable properties of market-clearing mechanisms are:

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- **Revenue adequacy** refers to a condition under which the market operator never incurs a financial deficit.

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- **Revenue adequacy** refers to a condition under which the market operator never incurs a financial deficit.
- In other words, the total payment that the market operator receives from demands is always higher than or equal to her total payment to generators, curtailed loads, transmission operator, etc.

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- **Revenue adequacy** refers to a condition under which the market operator never incurs a financial deficit.
- In other words, the total payment that the market operator receives from demands is always higher than or equal to her total payment to generators, curtailed loads, transmission operator, etc.
- As a specific status of revenue adequacy, the market is “**budget balanced**” if the market operator has neither financial deficit nor excess.

Discussion

Question:

Is there any market-clearing mechanism ensuring all four properties?

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Answer:

No!



- Based on Hurwicz theorem (also known as “impossibility theorem”) [1]-[2], no mechanism is capable of achieving all those properties at the same time!
- We have to find a “trade-off” among properties achieved and those lost.

[1] L. Hurwicz, “On Informationally Decentralized Systems” in *Decision and Organization*, edited by C.B. McGuire and R. Radner, Amsterdam, 1972.

[2] R. Myerson and M. A. Satterthwaite, “Efficient mechanisms for bilateral trading,” *Journal of Economic Theory*, vol. 28, pp. 265–281, 1983.

Discussion

Recall that market-clearing models determine the nodal market-clearing prices (LMPs) based on dual variable of nodal power balance equalities. Let's refer this pricing method to as “**LMP-based market mechanism**”.

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Equilibrium

For each generator:

Maximize profit

subject to production limits

For each demand:

Maximize utility

subject to consumption limits

Power balance

Optimization

Maximize market's social welfare

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Question:

Which properties are ensured in the LMP-based market mechanism?

Properties of LMP-based market mechanism



“Incentive compatibility” ensured?

“Market efficiency” ensured?

“Revenue adequacy” ensured?

“Cost recovery” ensured?

Properties of LMP-based market mechanism



“Incentive compatibility” ensured?

No! A market player (the so-called “strategic” player) may exercise “market power” by not trustfully offering in terms of price and/or quantity!



“Market efficiency” ensured?

“Revenue adequacy” ensured?

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Properties of LMP-based market mechanism



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“Market efficiency” ensured?

No! in the sense that if “market power” is exercised, the market’s social welfare will be decreased.



“Revenue adequacy” ensured?

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“Revenue adequacy” ensured?

Yes! Proof as an exercise; see the next slide!



“Cost recovery” ensured?

Yes! Proof as an exercise; see the next slide!



Exercise 1

Provide a mathematical proof that the LMP-based market mechanism ensures “revenue adequacy” and even “budget balancedness”.

Guide:

Step 1- Consider the nodal power balance equality for bus n .

Step 2- Multiply each term within the equality of Step 1 by the LMP at that bus.

Step 3- Consider the summation of all equalities obtained in Step 2 for all buses. What does the resulting equality mean?

Exercise 2

Provide a mathematical proof that the LMP-based market mechanism ensures “cost recovery” for all market players.

Guide:

Step 1- Consider the equilibrium form of the market-clearing problem.

Step 2- For each generator's optimization problem, derive the corresponding “strong duality” condition, which enforces the equality of objective function of primal and dual problems at the optimal point. The primal objective function is generator's profit. Check the terms within the dual objective function -- are they all non-negative? If so, what does it mean?

Step 3- Similar to Step 2, investigate the cost recovery for elastic demands and transmission system operator using the equilibrium problem in Step 1.

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Question: If the lower bound for the generation level of a generator is a positive (non-zero) value, can we still ensure cost recovery?

Unit commitment

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- etc.

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Some references explaining the formulation of UC constraints:

[1] M. Carrión and J. M. Arroyo, "A computationally efficient mixed-integer linear formulation for the thermal unit commitment problem," *IEEE Transactions on power systems*, vol. 21, no. 3, pp. 1371-1378, 2006.

[2] J. Ostrowski, M. F. Anjos, A. Vannelli, "Tight mixed integer linear programming formulations for the unit commitment problem," *IEEE Transactions on Power Systems*, vol. 27, no. 1, pp. 39-46, 2012.

[3] M. F. Anjos and A. J. Conejo, "Unit commitment in electric energy systems," *Foundations and Trends® in Electric Energy Systems*, vol. 1 no. 4, pp. 220-310, 2017.

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- Reference [1] uses **one** type of binaries only (on/off status of generators) to model all UC constraints.
- Reference [2] models the same UC constraints as in [1], but using **three** types of binaries (on/off, start-up, and shut-down status of generators).
- Both models obtain the same results, but the model in [2] is computationally **faster** than the one in [1]! Any idea why?

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One more reference!

If you would like to “relax” binary variables of UC constraints in a “tight” way, e.g., for capacity expansion planning studies:

[4] B. Hua, R. Baldick, and J. Wang, “Representing operational flexibility in generation expansion planning through convex relaxation of unit commitment,” *IEEE Transactions on power systems*, vol. 33, no. 2, pp. 2272–2281, 2018.

Discussion

Note: The unit commitment problem is a mixed-integer linear problem (MILP), and we still need dual variables to derive market prices!

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Question: How to derive dual variables in a MILP? Is there any mathematical challenge?

Some seminal papers about duality theory for integer programming:

- [1] E.L. Johnson, "Cyclic groups, cutting planes and shortest paths", *Mathematical programming*, 1973.
- [2] L. A. Wolsey, "Integer programming duality: Price functions and sensitivity analysis," *Mathematical Programming*, vol. 20, no. 1, pp. 173-195, 1981.
- [3] A. C. Williams, "Marginal values in mixed integer linear programming," *Mathematical Programming*, vol. 44, no. 1-3, pp. 67-75, 1989.

Discussion

Questions: When UC constraints are included,

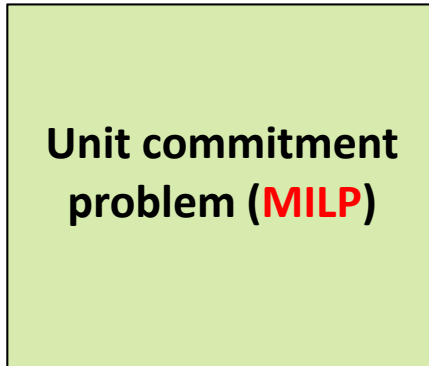
1- How to derive “market prices”?

2- Does the LMP-based market-clearing mechanism still guarantee achieving “cost recovery” for all market players?

Pricing with unit commitment constraints

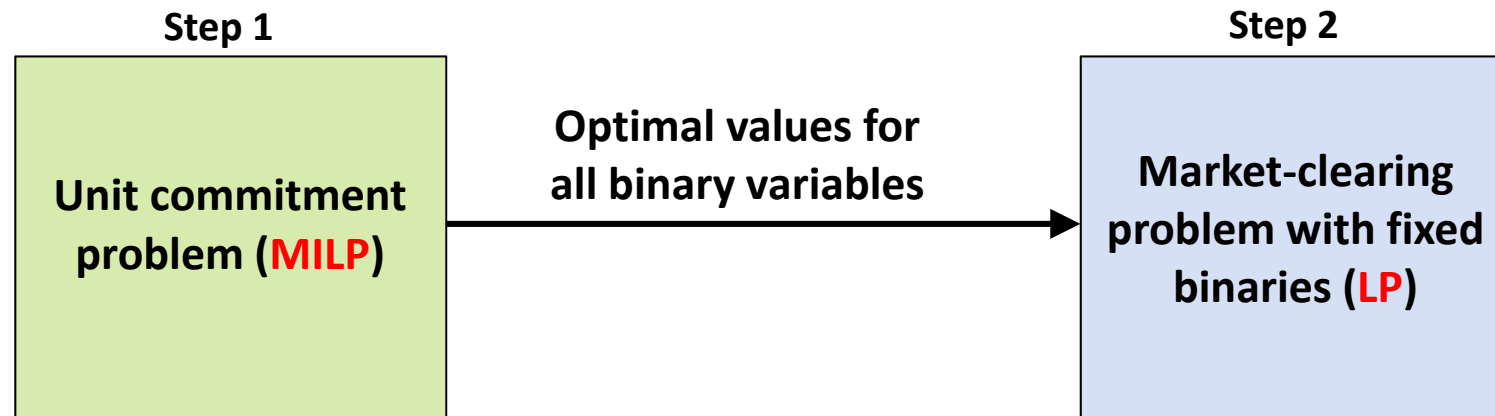


Step 1



- **Step 1:** Solve the unit commitment problem, and obtain the optimal values for all binary variables

Pricing with unit commitment constraints



- **Step 1:** Solve the unit commitment problem, and obtain the optimal values for all binary variables
- **Step 2:** Solve the same problem while binary variables are replaced by their optimal values (0 or 1) obtained in Step 1. This results in a linear problem (LP), so determine the LMPs based on dual variables.

Step 1: unit commitment problem

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Let's add the following UC constraints to the market-clearing problem:

- The minimum production level of generators,
- Start-up cost of generators.

Step 1: unit commitment problem

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subject to:

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Step 2: LP with fixed binaries

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$$\text{Maximize}_{p_{g,t}^G, s_{g,t}^G, p_{d,t}^D, \theta_{n,t}} \sum_{d,t} U_{d,t} p_{d,t}^D - \sum_{g,t} [s_{g,t}^G + (C_g p_{g,t}^G)]$$

subject to:

$$0 \leq p_{d,t}^D \leq \overline{P}_{d,t}^D \quad \forall d, \forall t$$

$$\underline{P}_{g,t}^G u_{g,t}^{G,\text{fixed}} \leq p_{g,t}^G \leq \overline{P}_g^G u_{g,t}^{G,\text{fixed}} \quad \forall g, \forall t$$

$$\sum_{d \in \Psi_n} p_{d,t}^D + \sum_{m \in \Omega_n} B_{n,m} (\theta_{n,t} - \theta_{m,t}) - \sum_{g \in \Psi_n} p_{g,t}^G = 0 \quad : \lambda_{n,t} \quad \forall n, \forall t$$

$$-F_{n,m} \leq B_{n,m} (\theta_{n,t} - \theta_{m,t}) \leq F_{n,m} \quad \forall n, \forall m \in \Omega_n, \forall t$$

$$\theta_{(n=\text{ref}),t} = 0 \quad \forall t$$

$$0 \leq s_{g,t}^G \leq C_g^{\text{SU}} (u_{g,t}^{G,\text{fixed}} - u_{g,t-1}^{G,\text{fixed}}) \quad \forall g, \forall t$$

Step 2: LP with fixed binaries

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Questions:

Does the start-up cost affect market prices?

Step 2: LP with fixed binaries

$$\text{Maximize}_{p_{g,t}^G, s_{g,t}^G, p_{d,t}^D, \theta_{n,t}} \sum_{d,t} U_{d,t} p_{d,t}^D - \sum_{g,t} [s_{g,t}^G + (C_g p_{g,t}^G)]$$

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By fixing the values of binaries, the optimal values of the start-up cost of generators are somehow given! Their start-up cost variable looks like a parameter in the objective function, and can be eliminated!

Step 2: LP with fixed binaries

$$\text{Maximize}_{p_{g,t}^G, s_{g,t}^G, p_{d,t}^D, \theta_{n,t}} \sum_{d,t} U_{d,t} p_{d,t}^D - \sum_{g,t} [s_{g,t}^G + (C_g p_{g,t}^G)]$$

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By fixing the values of binaries, the optimal values of the start-up cost of generators are somehow given! Their start-up cost variable looks like a parameter in the objective function, and can be eliminated!



The start-up cost does NOT contribute to the market price formation, because that cost is not reflected in the dual variable!

Step 2: LP with fixed binaries

$$\text{Maximize}_{p_{g,t}^G, s_{g,t}^G, p_{d,t}^D, \theta_{n,t}} \sum_{d,t} U_{d,t} p_{d,t}^D - \sum_{g,t} [s_{g,t}^G + (C_g p_{g,t}^G)]$$

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$$-F_{n,m} \leq B_{n,m} (\theta_{n,t} - \theta_{m,t}) \leq F_{n,m} \quad \forall n, \forall m \in \Omega_n, \forall t$$

This means that the “cost recovery” of generators **cannot** be ensured anymore, because market prices do not support their start-up costs!

By fixing the values of binaries, the optimal values of the start-up cost of generators are somehow given! Their start-up cost variable looks like a parameter in the objective function, and can be eliminated!

The start-up cost does NOT contribute to the market price formation, because that cost is not reflected in the dual variable!

Step 2: LP with fixed binaries

- In current practice of electricity markets in the U.S., market operators use “**uplift mechanisms**” [1]-[2] to restore “cost recovery”, but at the cost of inefficiency and sub-optimality!
- The negative profit of each generator is compensated by an “ex-post” side payment, i.e., “uplift payment”.
- The loads will be eventually charged due to uplift payments!

[1] R. O’Neill et al., “Efficient market-clearing prices in markets with nonconvexities,” *European Journal of Operational Research*, vol. 164, pp. 269–285, Jul. 2005.

[2] W. W. Hogan and B. J. Ring, “On minimum-uplift pricing for electricity markets,” *Harvard Working Paper*, 2003. Available: https://scholar.harvard.edu/whogan/files/minuplift_031903.pdf

Step 2: LP with fixed binaries

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Discussion:

What is the current practice in the European electricity markets?

Guide: Read about “block orders” and “paradoxically rejected blocks”

Exercise 3

There is another market-clearing mechanism called “Vickrey–Clarke–Groves (VCG)”, which pays/charges each generator/demand based on its value/cost to the market (see, e.g., references [1]-[2], in the next slide).

Consider a market with three generators G1, G2, G3 and two demands D1 and D2. The market-clearing outcomes in VCG are identical to those in the LMP-based market design; the only difference is “pricing” (but NOT production and consumption levels)!

For example, to calculate the payment to G1 under VCG, the market operator clears the market twice: one time including G1, and another time excluding G1. The payment to G1 is:

$$\text{Payment to G1} = A - B$$

where term A is calculated based on clearing outcomes when G1 exists in the market:

$$A = [\text{total utility of demands D1 and D2}] - [\text{total cost of generators G2 and G3, but not G1}]$$

Similarly, term B is calculated based on market-clearing outcomes when G1 is absent:

$$B = [\text{total utility of demands D1 and D2}] - [\text{total cost of generators G2 and G3}]$$

Exercise 3

Question:

- Does VCG mechanism ensure “incentive compatibility”, “market efficiency” and “cost recovery”?

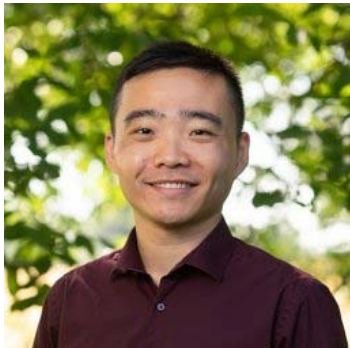
Note: The VCG mechanism does not necessarily achieve “revenue adequacy” for the market, which is its main drawback! There are several versions of improved VCG in the literature, seeking to reduce (but not necessarily eliminate) the budget deficit.

[1] Y. Xu and S. H. Low, “An efficient and incentive compatible mechanism for wholesale electricity markets,” *IEEE Transactions on Smart Grid*, vol. 8, no. 1, pp. 128-138, 2017.

[2] B. F. Hobbs, M. H. Rothkopf, L. C. Hyde, and R. P. O'Neill, “Evaluation of a truthful revelation auction in the context of energy markets with non-concave benefits,” *Journal of Regulatory Economics*, vol. 18, no. 1, pp. 5-32, 2000.

Next Lecturer (on January 7, 2021)

Cooperative Game Theory in Energy Systems



Dr. Liyang Han

- Postdoc (DTU), July 2020-ongoing
- PhD (University of Oxford), 2020
- MSc (Stanford University), 2014
- BSc (Zhejiang University), 2012

His research interests include the integration of distributed energy resources (DER) and Peer-to-Peer energy markets. Liyang's PhD thesis was about the application of cooperative game theory in constructing energy coalitions in the distribution network. His current research focus at DTU is on data market design for energy systems.

Thanks for your attention!

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